

,Princeton University
Mathematical Intermediate Microeconomics (ECO 310)
Midterm Spring 2023

Instructions: You have 80 minutes to complete this exam, plus 10 minutes to upload your answers. There are 6 multiple choice questions and 3 short answer questions. **In the Multiple Choice section, simply indicate the correct answer. In the Short Answer section, show your work and explain your answers for full credit!**

- The exam is open-book, and calculators are permitted. However, please work **alone**.
- **If you are unsure whether you're interpreting a question correctly, state your interpretation and proceed (or email me if you prefer).**

The honor code applies, as usual.

Q1: Multiple Choice (24 points, 4 each)

B C A A D D

1. Jake tells you that when comparing any two numbers a and b , he prefers a if and only if it's at least as large as $b + 1$. Jake is:

- (a) Rational
- (b) Irrational because these preferences are incomplete
- (c) Irrational because these preferences are intransitive
- (d) Two of these are correct

Answer: B. These are incomplete, eg if the numbers are 1 and 1.5, he is unable to compare them (it's not true that $1 \geq 1.5 + 1$, and it's not true that $1.5 \geq 1 + 1$). But they are transitive: if $a \succsim b$ and $b \succsim c$, it means $a \geq b + 1$ and $b \geq c + 1$; together this implies $a \geq c + 2 > c + 1$, so $a \succsim c$.

2. A consumer's preferences over good 1 and good 2 are described by $u(x_1, x_2) = x_1^{\frac{1}{2}} x_2$. If this consumer behaves optimally, he will:

- (a) Spend $\frac{1}{3}$ of income on good 2.
- (b) Spend $\frac{1}{2}$ of income on good 2.
- (c) Spend $\frac{2}{3}$ of income on good 2.
- (d) None of these is correct.

Answer: C. Raising to the power $\frac{2}{3}$ to get exponents that sum to 1, these are equivalent to preferences represented by utility $x_1^{\frac{1}{3}} x_2^{\frac{2}{3}}$; so by the Cobb Douglas magic rule this consumer optimally spends $\frac{2}{3}$ of their income on good 2

3. A consumer initially buys 5 oranges and 10 apples. Then an apple tax increases the price per apple by \$0.50, but the consumer is at the same time given a \$5 fruit subsidy. Which of the following best describes this situation?

- (a) The consumer is most likely better off now (with the tax and subsidy) than he was originally.
- (b) The consumer is most likely worse off now (with the tax and subsidy) than he was originally.
- (c) The consumer is definitely indifferent between his original and current situations.

Answer: A. With the apple tax, the original bundle now costs \$5 more than it did before. So if the consumer gets a \$5 fruit subsidy, they are able to get exactly their original bundle. But there is likely a more efficient way to spend money than by sticking with the original bundle, so this consumer can likely do better.

4. A consumer gets utility $u(L, R) = \min\{L, R\}$ from L left shoes and R right shoes. If $m = 10$, $p_L = 1$, and p_R increases from 1 to 4, the change in demand for right shoes due to the substitution effect is:

- (a) 0

- (b) -1
- (c) -2
- (d) -3
- (e) None of these is correct

Answer: A. With perfect complements, if compensated to get something as good as the original bundle, the most efficient way to do this is to get exactly the original bundle (you still want $L=R$, so to maintain utility you need to buy the same # pairs of shoes as before). No reoptimization means no substitution effect.

5. At the end of Lecture 10, we saw the following “paradox”: many people prefer saving 200 of 600 people for sure, to a risky option with a $\frac{1}{3}$ chance of saving all 600 people and a $\frac{2}{3}$ chance of saving nobody. But they prefer a risky option with a $\frac{1}{3}$ chance of killing nobody and a $\frac{2}{3}$ chance of killing everybody, to an option that kills 400 people for sure.

If people care about the # people alive in the end, then

- (a) This is inconsistent with EU maximization
- (b) This violates consequentialism
- (c) This violates independence
- (d) Two of these are true
- (e) All of these are true

Answer: D. When it’s phrased as saving lives, they prefer having 200 people definitely alive, to a risky option with 200 alive in expectation. But when it’s phrased as killing people, they prefer a risky option that leaves 200 alive in expectation, over a safe option that leaves 200 alive (400 killed) for sure. This violates consequentialism, since the way we frame the lottery affects the choice, people do not simply care about final distributions over outcomes. But there is no violation of independence (here we’re talking about the same two lotteries described differently). Of course this is inconsistent with EU maximization, which inherently cares only about probability distributions over final outcomes.

6. Which of the following would a risk-averse consumer **definitely** prefer to a 50-50 \$0 or \$200 gamble:

- (a) \$90 lump-sum
- (b) \$110 lump-sum
- (c) A 50-50 \$50 or \$150 gamble
- (d) Two of a-c are definitely preferred
- (e) Three of a-c are definitely preferred

Answer: D. The gamble has an expected value of \$100, but with risk. So \$90 may or may not be better, it’s safe (good) but with a lower return (bad). \$110 is definitely better, higher value and less risk. Option C is also better, since this has the same expected value \$100, but bringing the outcomes in closer together reduces the risk.

Q2: Proofs and General Concepts (20 Points)

- a. (12) Assume that Melsi and Mavi (my dogs) are equally wealthy. Both are risk-averse, but Mavi is **more** risk-averse. Prove through the following steps that any time Mavi prefers accepting a risky bet (to rejecting it), so does Melsi:

- i. (6) Show that someone prefers accepting (rather than rejecting) a risky bet if and only if their certainty equivalent for the gamble exceeds their initial wealth.

Answer: accepting the bet implies that its EU, which equals $u(CE)$ by definition, exceeds $u(w_0)$; since utility increases with wealth, $u(CE) > u(w_0) \Rightarrow CE > w_0$. And conversely if $CE > w_0$, then $u(CE) > u(w_0)$ since u is increasing, while $EU = u(CE)$ by definition of the certainty equivalent; so $EU > u(w_0)$, and so this person prefers to take the bet since it yields a higher utility EU than sticking with initial wealth.

- ii. (6) If Mavi is more risk-averse than Melsi, then from Lecture 10, her risk premium is higher for every gamble. Prove that this implies that for a particular gamble, Melsi's certainty equivalent is higher than Mavi's.

Answer: The risk premium π is defined by $EU = u(E[W] - \pi)$, i.e. the amount of wealth π that the decision-maker would sacrifice to get a sure thing. We also have $EU = u(CE)$, and so comparing gives $E[W] - CE = \pi$. So for any gamble, Mavi's π is higher than Melsi's; since $E[W]$ is the same (they are facing the same gamble), it follows that Mavi's CE is lower than Melsi's.

Since they are initially equally wealthy, this means by (a) that whenever Mavi's CE exceeds initial wealth w_0 and so she accepts the bet by (a), so does Melsi's.

- b. (8) You are working for a law firm that represents an elderly man who wants to leave his fortune to his dog. His evil family plans to have him committed if he is caught violating WARP. Previously, he consumed $(4, 2)$ when prices were $(1, 3)$. Prices are now $(3, 5)$. Which bundles (x_1, x_2) would get him committed if he purchased them now? (Just write out the inequality(ies) that must hold, no need to simplify).

He would get committed if he bought $(4, 2)$ over (x_1, x_2) when it costs more, and then now buys (x_1, x_2) over $(4, 2)$ when it costs more (at least one "more" meaning "strictly more"): i.e. if

$$\begin{aligned} (1, 3) \cdot (4, 2) &\geq (1, 3)(x_1, x_2) \Leftrightarrow 10 \geq x_1 + 3x_2 \\ \&(3, 5) \cdot (4, 2) &\leq (3, 5) \cdot (x_1, x_2) \Leftrightarrow 22 \leq 3x_1 + 5x_2 \end{aligned}$$

(With at least one of these inequalities strict).

Q3: Demand (37 Points)

Harry's preferences over netflix hours (N) and cable hours (C) are represented by

$$u(N, C) = C^\alpha + 2CN$$

where $\alpha > 0$ is a parameter.

- a. (8) Show that if $\alpha \leq 2$, then these preferences are convex.

Answer: here $\frac{MU_C}{MU_N} = \frac{\alpha C^{\alpha-1} + 2N}{2C} = \frac{\alpha}{2} C^{\alpha-2} + \frac{N}{C}$. We would like this to fall as $C \uparrow$ and $N \downarrow$. First of all, increasing C and decreasing N reduces $\frac{N}{C}$ (numerator falls, denominator rises). And if $\alpha \leq 2$, then increasing C also weakly reduces $C^{\alpha-2}$, as its derivative in C is $(\alpha - 2)C^{\alpha-3}$, which is ≤ 0 for $\alpha \leq 2$. So convexity holds in this range.

- b. (5) Write the equation for Harry's indifference curve through the point $(1, 1)$. (No need to simplify).

Answer: this is all points (C, N) yielding the same utility as $(1, 1)$, i.e.

$$C^\alpha + 2CN = 1^\alpha + 2 \cdot 1 \cdot 1 = 3$$

- c. (12) Find Harry's (Marshallian) demand function if $\alpha = 2$, assuming income m and prices p_C and p_N per hour of cable and netflix.

Answer: Answer: he wants to spend all his money and allocate it optimally between goods. The optimal allocation condition is

$$\frac{MU_C}{MU_N} = \frac{p_C}{p_N} \Rightarrow 1 + \frac{N}{C} = \frac{p_C}{p_N} \Rightarrow \frac{N}{C} = \frac{p_C - p_N}{p_N} \Rightarrow p_N N = (p_C - p_N)C$$

(This is nonnegative for N so long as $p_C \geq p_N$, assume this for now). Now substitute this into the budget line, $p_C C + p_N N = m$, to get

$$p_C C + (p_C - p_N)C = m \Rightarrow (2p_C - p_N)C = m \Rightarrow C = \frac{m}{2p_C - p_N}$$

Then substitute this back into the bang per buck condition to get

$$N = \frac{1}{p_N} \frac{p_C - p_N}{2p_C - p_N} m$$

Finally, checking nonnegativity conditions: if $p_C \geq p_N$, this all works. If $p_C < p_N$, then bang per buck gave a negative quantity of N , so go for the next best thing, $N = 0$. Substituting this into the budget constraint then gives $C = \frac{m}{p_C}$.

- d. (12) Finally, suppose Harry's limited resource is time rather than money: Netflix and cable are both offered free of charge in his dorm room, but he has limited time available to watch them. (i) What is his budget constraint if he has T hours available? (ii) Again assuming that $\alpha = 2$, how much time T does he need to get utility u^* ?

Answer to (i) his budget constraint is now $C + N \leq T$ (comparing to the one in (d), effective prices are $p_C = 1$ and $p_N = 1$). For (ii), this is asking about the expenditure function: to get utility u^* in the most efficient way, he firstly equates bang per buck across goods, so

$$\frac{MU_C}{MU_N} = \frac{p_C}{p_N} = 1 \Rightarrow 1 + \frac{N}{C} = 1 \Rightarrow N = 0$$

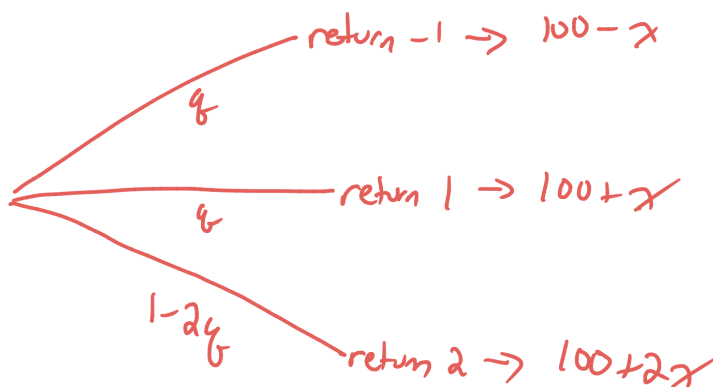
So he will not buy any netflix. Then his actual utility is $[C^\alpha + 2CN]_{\alpha=2, N=0} = C^2$. So to get utility u^* in the most efficient way possible, he sets $C^2 = u^* \Rightarrow C = \sqrt{u^*}$, along with $N = 0$. Therefore the time needed for this is $T = \sqrt{u^*}$ hours.

Q4: Expected Utility (31 Points)

Suppose you are *risk-averse*, have \$100 (no wealth besides this), and are considering investing some of your \$100 in an asset with 3 possible return rates: $r_1 = -1$ (which happens with probability $q \leq \frac{1}{2}$), $r_2 = 1$ (which also happens with probability $q \leq \frac{1}{2}$), and $r_3 = 2$ (this happens whenever the other two possible return rates do not). Your other option is to hide the money in a sock drawer. Recall that \$ x invested at return rate r **changes** your wealth by \$ rx .

- a. (12) Write out your expected utility from investing \$ x of your \$100 in this asset, and keeping the remaining \$ $100-x$ in a sock drawer. (Hint: consider drawing a probability tree, with one branch for each way the uncertainty can resolve; at the end of each branch figure out your final utility if this happens.)

Answer: following the hint, 3 return rates means 3 ways the uncertainty can resolve, so the probability tree has 3 branches. On the branch with return rate r , final wealth = initial wealth + change in wealth = $100 + rx$. So here is the probability tree:



and then EU is the probability-weighted average of the final utility levels:

$$EU = qu(100 - x) + qu(100 + x) + (1 - 2q)u(100 + 2x)$$

Comment: amazingly few people got this, I wish more had followed the hint.

- b. (14) If you can invest any amount x between \$0 and \$100 (inclusive, decimals allowed),
- (5) Explain: at $q = 0$, you should invest your entire \$100 in the asset.
Answer: at $q = 0$, EU above simplifies to simply $u(100 + 2x)$; this is increasing in x , so you would like to make x as large as possible.
 - (9) Find all values of q (if any) for which it is optimal to keep all money in your sock drawer.
Answer: you want to keep all money in the sock drawer, i.e. invest $x = 0$, if either $EU'(0) = 0$ (your FOC for the optimal x is maximized exactly at $x = 0$), or if $EU'(0) \leq 0$ (even with $x = 0$, EU is $\downarrow$ in x , implying that the more you invest the worse it is; this is where you'd invest a negative x if possible, but since this isn't allowed, you go for the best corner $x = 0$). From (a),

$$EU'(x) = -qu'(100 - x) + qu'(100 + x) + 2(1 - 2q)u'(100 + 2x)$$

Evaluating at $x = 0$,

$$EU'(0) = -qu'(100) + qu'(100) + 2(1 - 2q)u'(100) = 2(1 - 2q)u'(100)$$

This equals zero at $q = \frac{1}{2}$, and is strictly positive if $q < \frac{1}{2}$ (and we were given the range $q \leq \frac{1}{2}$). So this only happens at $q = \frac{1}{2}$.

- c. (5) If your only options were to invest \$20 or \$80, how would you figure out which is better?
Answer: you would pick \$20 if $EU(20) > EU(80)$, invest \$80 if the reverse inequality holds, and are indifferent if these two EU expressions are equal. (Where $EU(20)$ is what you get by evaluating the EU expression in (a) at $x = 20$, and similarly for $EU(80)$).